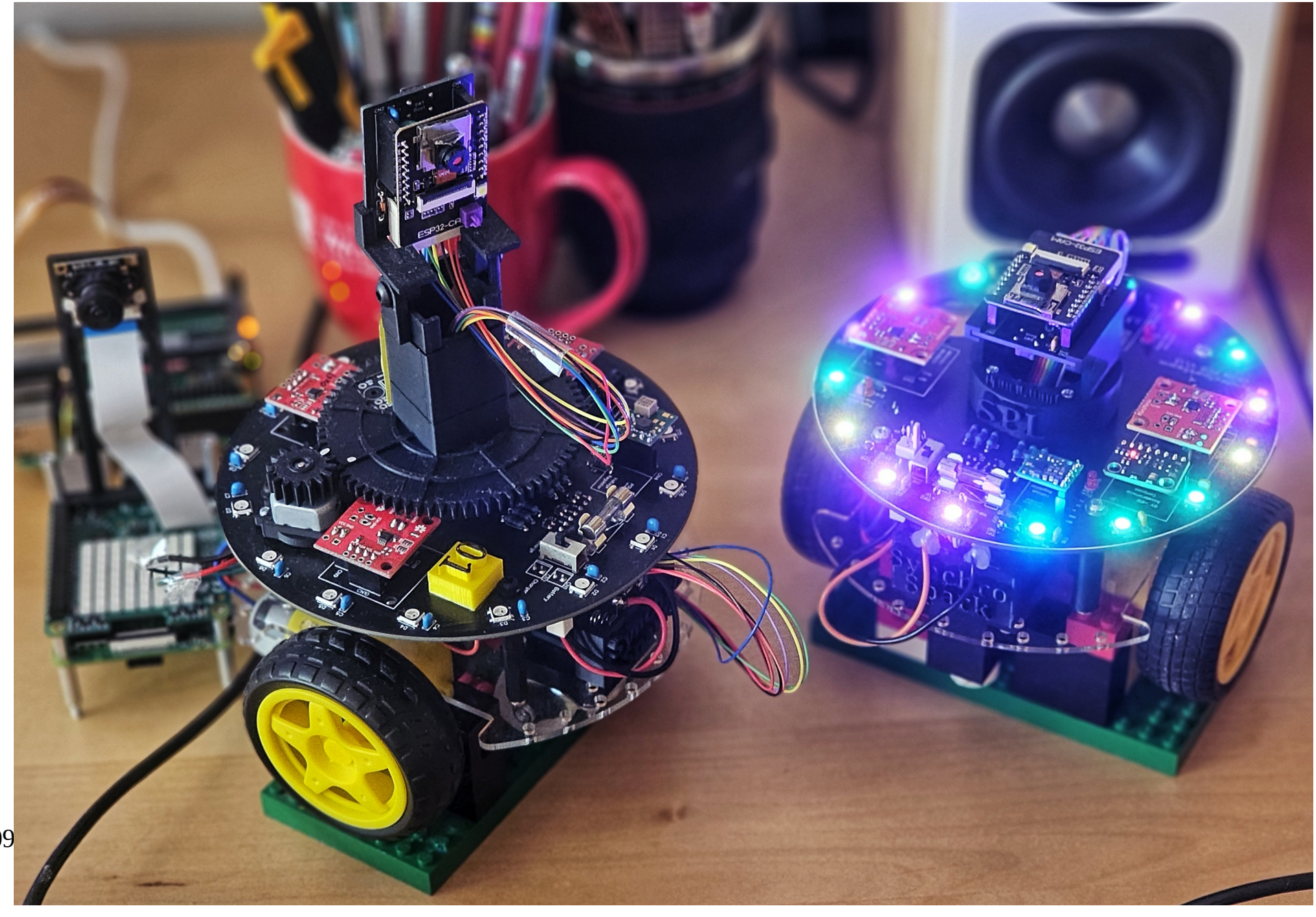
A Cost-Effective and Scalable Hybrid-Reality Testbed for Robot Networks Integrating Physical Agents and Digital Twins

Vedant_Agarwal Ismet_Atalar Andy_Li
Ahmet_Sekercioglu Warwick_Stone

SparkNodes



Physical Network



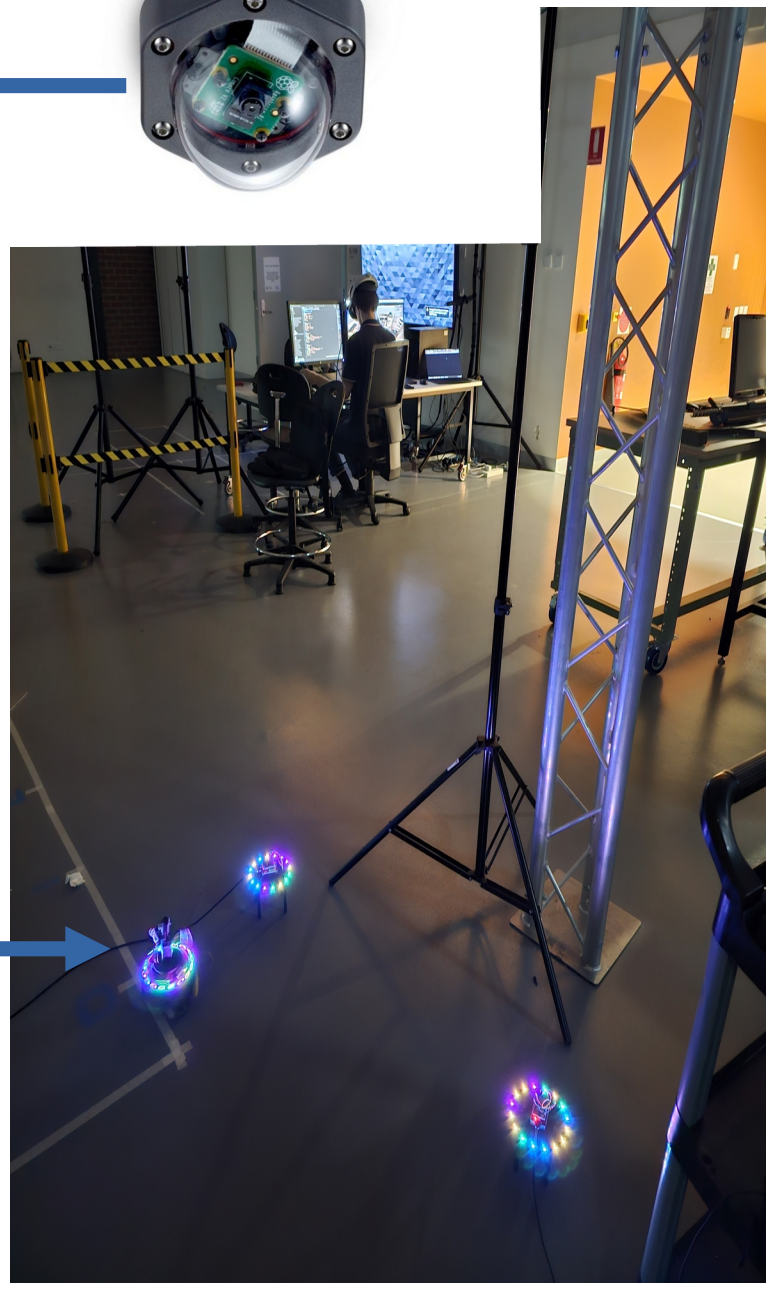
tracking



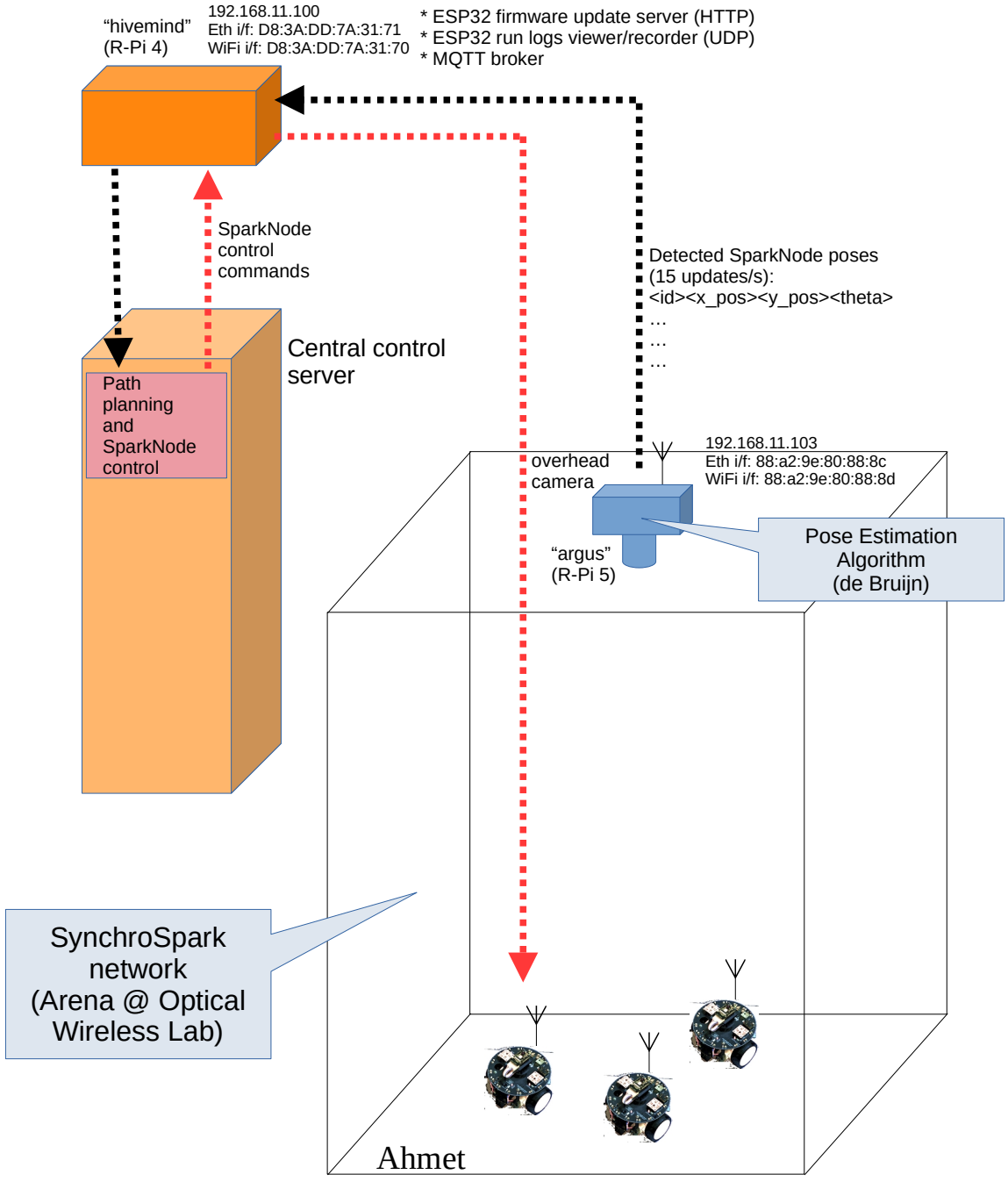
Formation control algorithm

```
# Drive forward at speed 32 for 5000 milliseconds (5 seconds)
mosquitto_pub -h <broker_ip> -t "arena/sparknode01/cmd" -m "drive forward 32 5000"
# Turn right at speed 25 for 1500 milliseconds
mosquitto_pub -h <broker_ip> -t "arena/sparknode01/cmd" -m "turn right 25 1500"
# Slow reverse movement at speed 15 for 2000ms
mosquitto_pub -h <broker_ip> -t "arena/sparknode01/cmd" -m "drive reverse 15 2000"
# Fast turn at speed 45 for 800ms
mosquitto_pub -h <broker_ip> -t "arena/sparknode01/cmd" -m "turn left 45 800"
# Emergency stop
mosquitto_pub -h <broker_ip> -t "arena/sparknode01/cmd" -m "stop"
# Broadcast command to all sparknodes
mosquitto_pub -h <broker_ip> -t "arena/all/cmd" -m "stop"
```

Commands over MQTT

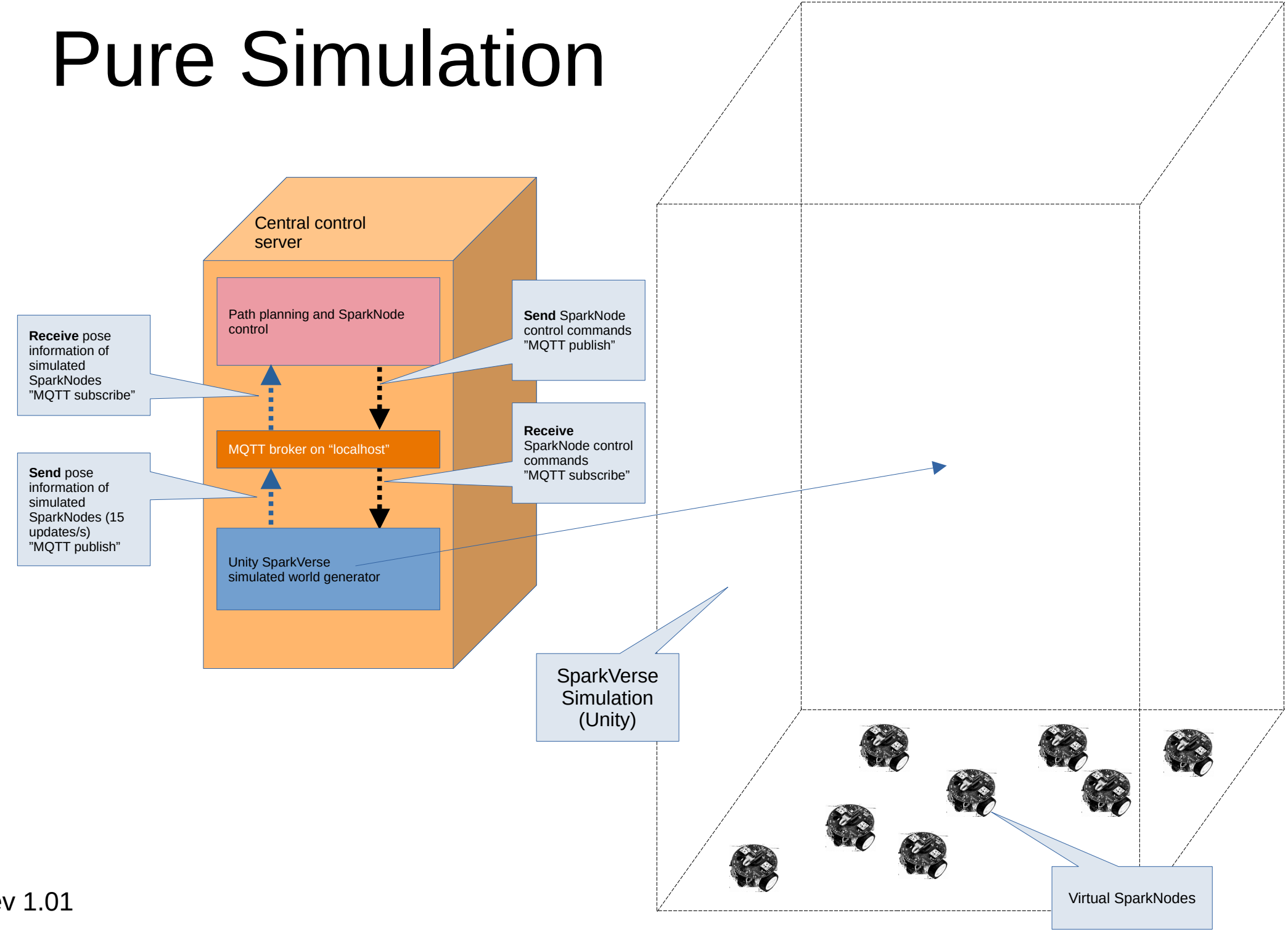


Physical Network



Rev 1.01

Pure Simulation



Rev 1.01

Hybrid System

